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idea. It was the design of an updated cotton stripper machine and it transformed him into a successful innovator. He invented the machine in 1991-92, perfecting it only in 1998 and it is now being used by nearly 100 cotton gins that process a small local seed, kala. The machine has brought down the cost of stripping cotton from around Rs 1 per kg to Rs 1 for 20 kg, ensuring not only quick returns for farmers but also improving the quality of processing.

The main components of the machine, which costs Rs 2 lakh, are a metallic wire and a brush that work together to pluck out cotton from the cottonball. Patel began research in 1990 when he found, on a visit to his village Trent, near Ahmedabad, that the cotton gin owners had requested a textile research organisation to invent a machine to strip cotton. Hailing from a farming family which also grew cotton in a small way, he was familiar with the recurring expenses and production delays. He requested a cooperative gin to give him funds to conduct research.

Patel managed to collect Rs 1 lakh with the help of the management as well as some friends and began research in a rented space, working simultaneously at the Asarwa Mill which chipped in by virtually freeing him from daily attendance. It took Patel nearly two years to develop the machine but the first 13 models he sold developed a snag. Furious cotton gin owners returned the

machines causing him a loss of Rs 20 lakh. But Patel continued his research and even asked his college-going sons Kirit and Nitin to drop out and join him. Srishti, an NGO that supports grassroots innovators, gave him Rs 2 lakh for research.

Success finally came in 1998 when he removed the snag. Confident of his dream machine, he resigned and launched a company—Chetak Industries at Viramgam which today, along with its two sister concerns, has a turnover of over Rs 4 crore. "The machine has had a multiple impact on the cotton ginning industry and has created a win-win situation for both the farmers and the cotton processors," says Patel. For Prabhudas Thakkar, a cotton gin owner, "the machine has made the business of stripping cotton both easy and lucrative". Where the manual removal of cotton from the pod used to cost between Rs 10 and Rs 15 per 20 kg, with the new and improved cotton stripper, the costs have dipped to only a rupee.

After getting both an Indian and an American patent, Patel and his sons have now set their eyes on developing a saline resistant brick which will be useful in large tracts of Saurashtra and north Gujarat. Though they have developed a brick, it is heavy and the trio is now working towards making it lighter. Knowing Patel, the day may not be far when the cotton stripper will become a brick builder. ■



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I want my grandchildren to take to technological research as well but I don't want them to be degreeless scientists like me.

MANSUKHBHAI PATEL,
Innovator-entrepreneur

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SHAILESH RAVAL